Decentralized Learning in Online Queuing Systems

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Outline

- Introduction and model
- Existing results
- The case for cooperation
- Decentralized learning strategy

Introduction and model

Motivations

- traffic routing (road network, servers network)
- repeated congestion games
- cars (packets) taking same road (server) interfere

Learning in repeated games

Learning game environment (e.g. server/road quality)
Strategic interest (minimize own cost)

(Hart and Mas-Colell, 2000)

In repeated games, no internal regret strategies converge (in average) to the set of correlated equilibria.

$\rightarrow$ simple policies with strategic and learning guarantees

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Assumption: outcome of each round is independent of the past (except in learning of agents)

Carryover feature

Road traffic
independence of rounds

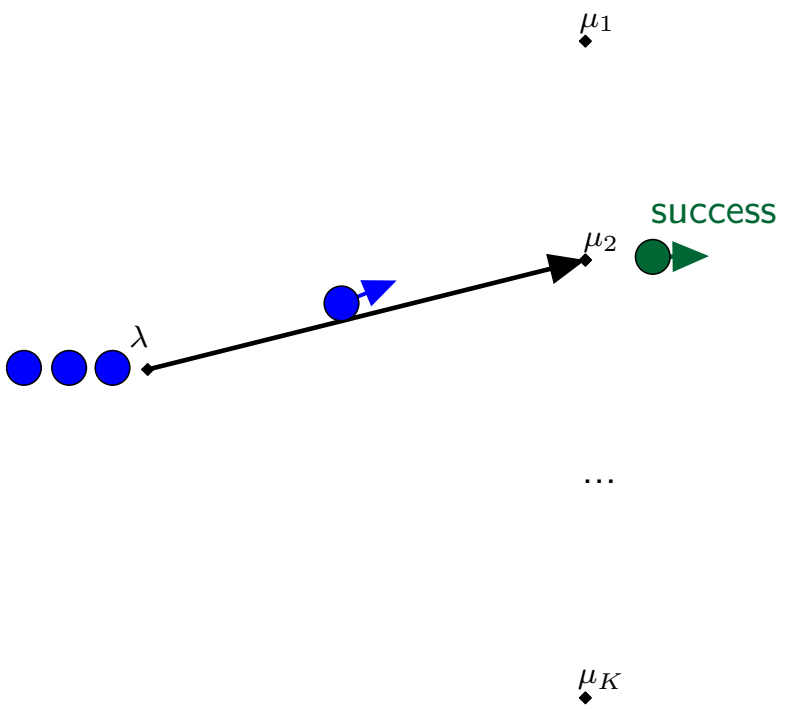
Second-by-second packet routing
Dropped packets have to be resent in next rounds

→ Learning in repeated games **with carryover**?

Model: single queue

At each $t = 1, \ldots, \infty$

- packet arrives with proba λ
- sends a packet to server $k \in [K]$
- server k clears with proba μ_k
- if fails $\rightarrow$ packet back in queue



$\rightarrow$ **multi-armed bandits approach**

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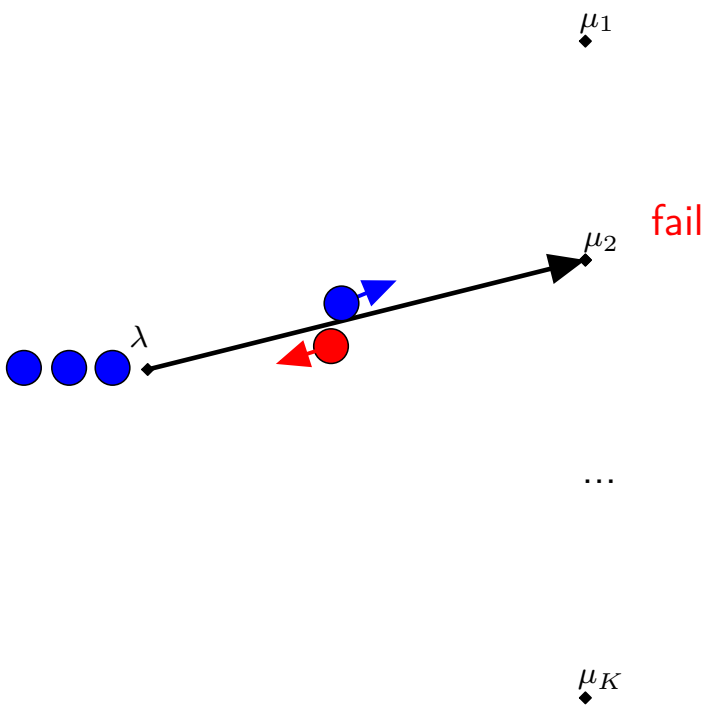
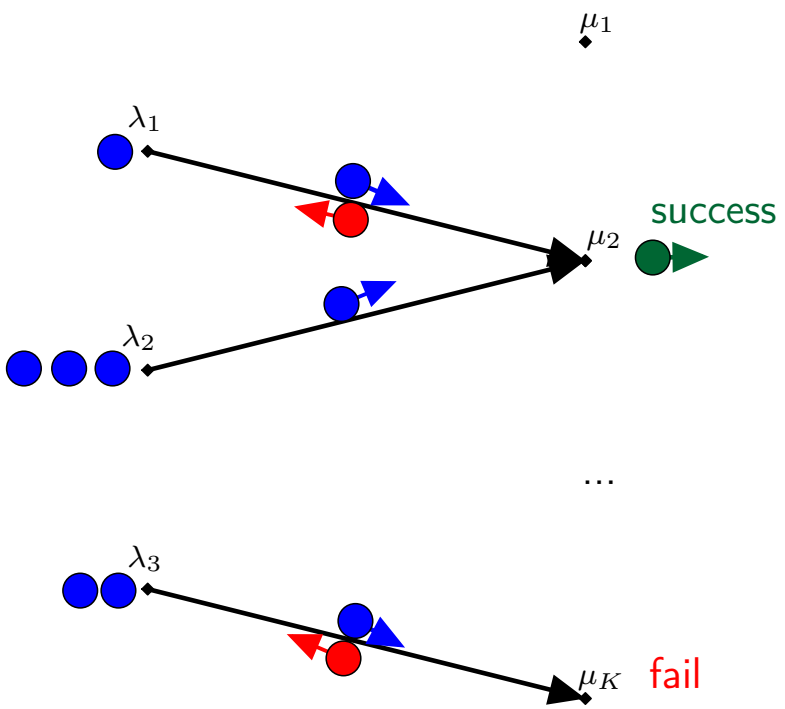


$\rightarrow$ **multi-armed bandits approach**

Model: multiple queues

- N queues ($N \leq K$)
- Heterogeneous arrival rates λ_i
- each queue chooses $a_t^i \in [K]$
- Server treats one packet at a time



related to multiplayer/competing bandits

Stability

Q_t^i number of packets in queue i at time t

A queue i is **stable** if for any r, there is a constant $C_r > 0$ such that

$$\mathbb{E}[(Q_t^i)^r] \leq C_r \qquad \forall t \in \mathbb{N}$$

System is stable if all queues are stable

Notation: $\lambda_{(1)} \geq \lambda_{(2)} \geq \ldots \geq \lambda_{(N)}$ and $\mu_{(1)} \geq \ldots \geq \mu_{(K)}$
Define slack

$$\eta = \max \left\{ \eta' \in \mathbb{R}_+ \mid \forall n \in [N], \eta' \sum_{i=1}^{n} \lambda_{(i)} \leq \sum_{i=1}^{n} \mu_{(i)} \right\}$$

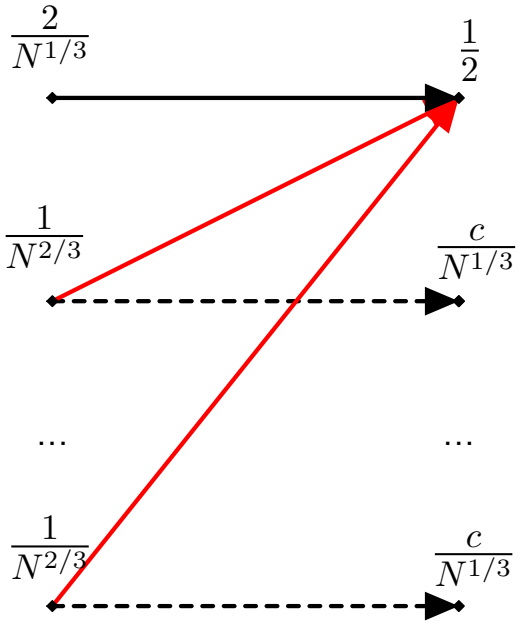
Centralized case: there is a stable strategy iff $\eta > 1$.

Goal: decentralized stable strategies for small η

Priority choice

A server can treat only one packet at a time.
Which packet to choose?

At random?
$\rightarrow$ unstable Nash equilibria with large η ($\gtrsim N^{1/3}$)



$N = K$
$\lambda_1 = \frac{2}{N^{1/3}}$ and $\lambda_i = \frac{1}{N^{2/3}}$ for all $i \geq 2$
$\mu_1 = \frac{1}{2}$ and $\lambda_i = \frac{c}{N^{1/3}}$ for all $i \geq 2$

queue 1 cannot clear

Priority choice

A server can treat only one packet at a time.
Which packet to choose?

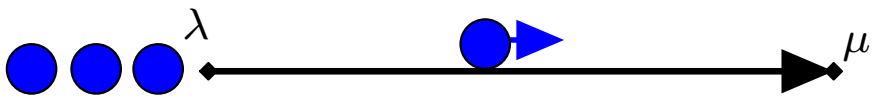
Treat oldest packet
$\rightarrow$ force better Nash equilibria
$\rightarrow$ carryover effect

if some queue accumulates packets $\rightarrow$ gets priority
bad performance for other queues on the long run $\rightarrow$ incites to cooperation

Existing results

Centralized case

Single queue, single server



Random walk (with frontier at 0)

- $\lambda < \mu$ $\rightarrow$ negative bias, stable
- $\lambda = \mu$ $\rightarrow$ no bias, queue size grows in $\sqrt{t}$
- $\lambda > \mu$ $\rightarrow$ positive bias, queue size in $(\lambda - \mu)t$

$\Longrightarrow$ centralized strategy stable iff $\eta > 1$

No regret strategies

Define regret

$$R_i(T) = \max_{k \in [K]} \sum_{t=1}^{T} \nu_t^i(k) - \sum_{t=1}^{T} \overbrace{\nu_t^i(a_t^i)}^{\text{1 if clears}}$$

Generally, learning agents selfishly minimize their regret

- no regret strategies known (eg EXP.3.P)
- used without strategic considerations
- converge in average to equilibria (without carryover effect)

No regret strategies

(Gaitonde and Tardos, 2020a)

If $\eta > 2$ and all queues follow a no regret strategy: $R_i(T) = o(T)$ w.h.p., then the system is stable.

$\rightarrow$ good when slack of the system is large

But minimizing regret is myopic and disregards long terms effect (carryover)
Can we do better?

Patient game

Define game $\mathcal{G} = ([N], (c_i)_{i=1}^n, \boldsymbol{\mu}, \boldsymbol{\lambda})$ with

Action Space: $p_i \in \mathcal{P}([K])$

Cost Function: All queues choose their server $a_t^i \sim p_i$ at each time step and

$$c_i(p_i, \boldsymbol{p_{-i}}) = \lim_{t \to +\infty} \frac{T_t^i}{t}$$

- T_t^i is the age of the oldest packet in queue i at time t
- this limit exists (deterministically)
- queue i is stable $\implies c_i(p_i, \boldsymbol{p_{-i}}) = 0$

Patient game

(Gaitonde and Tardos, 2020b)

If $\eta > \frac{e}{e-1}$, all Nash equilibria of $\mathcal{G}$ are stable.

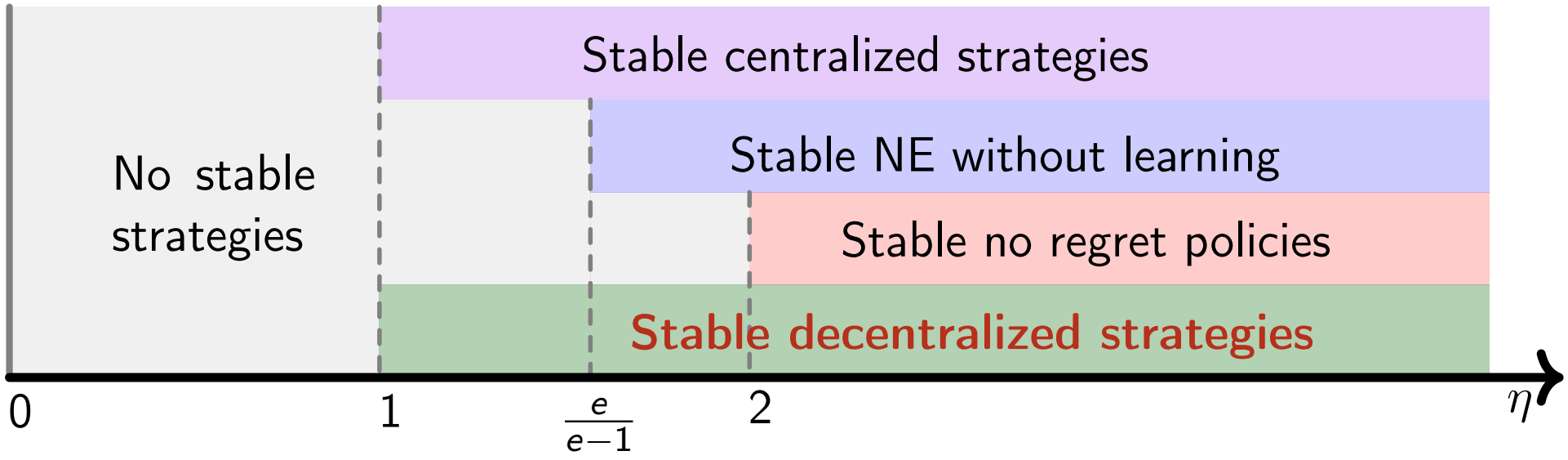
- long term considerations lead to stable strategies for $\eta \leq 2$
- $\frac{e}{e-1}$ is the *"price of anarchy"*
- reachable for learning agents?

Our contributions

1. cooperation required for stability of learning queues with $\eta \leq 2$
2. propose decentralized learning algorithm ADeQuA

Theorem

If $\eta > 1$ and all queues follow ADeQuA*, then the system is stable.*



The case for cooperation

A patient regret

Regret ignores the long term effects and leads to myopic decisions

$$\mathbb{E}[R_i^{\text{ext}}(T)] := \max_{p \in \mathcal{P}([K])} \sum_{t=1}^{T} \overbrace{\mathbb{E}_{\tilde{a}_t^i \sim p}[\nu_t^i(a_{1:t-1}^i, \tilde{a}_t^i)]}^{\text{reward we would get by playing } p \textbf{ for this round}} - \sum_{t=1}^{T} \overbrace{\mathbb{E}[\nu_t(a_{1:t}^i)]}^{\text{actual reward}}$$

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Policy regret instead takes into account the effects of an action on the environment

$$R_i^{\text{pol}}(T) := \max_{p \in \mathcal{P}([K])} \sum_{t=1}^{T} \overbrace{\mathbb{E}_{\tilde{a}_{1:t}^i \sim \otimes_{i=1}^t p}[\nu_t(\tilde{a}_{1:t}^i)]}^{\text{reward we would get if we were playing } p \textbf{ for all rounds}} - \sum_{t=1}^{T} \mathbb{E}[\nu_t(a_{1:t}^i)]$$

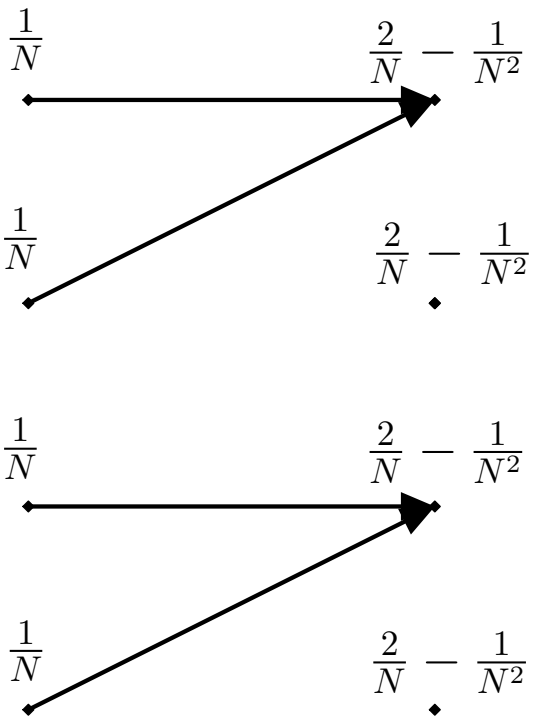
No policy regret strategies

Patient agents should minimize policy regret
Are no policy regret strategies more stable? **No**

There is an instance with $\eta = 2 - \mathcal{O}\left(\frac{1}{N}\right)$, s.t. each queue's policy regret is $o(T)$, but the system is not strongly stable.

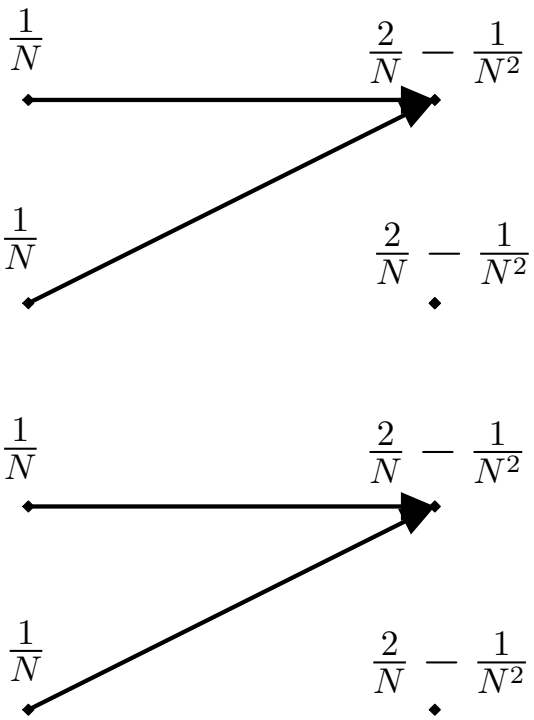
$\rightarrow$ no policy regret is *no better* than no regret

Counter Example



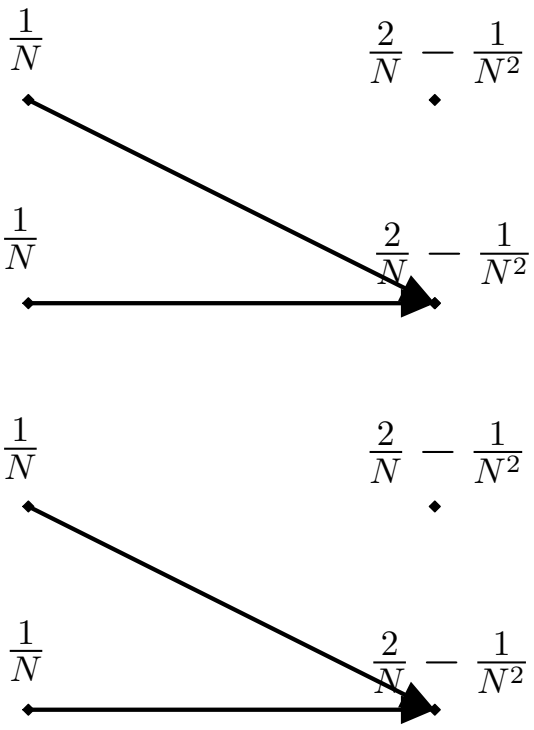
$\lambda_i = \frac{1}{N}$ for all i
$\mu_i = \frac{2}{N} - \frac{1}{N^2}$ for all i

Counter Example



First phase of length αT
Pairwise actions

Counter Example (first phase)



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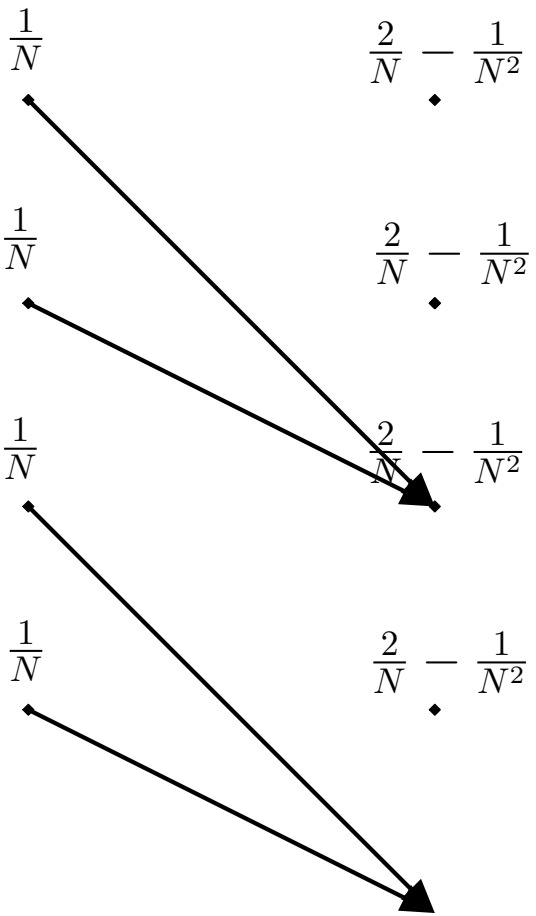
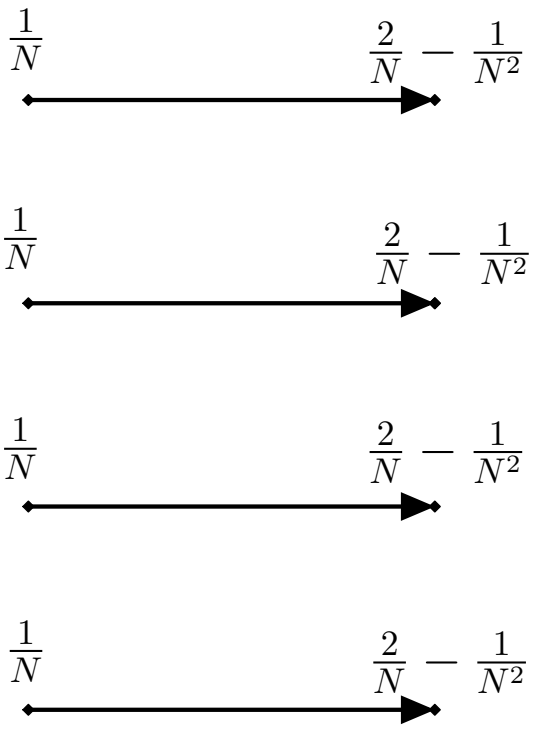


First phase of length αT
Pairwise actions

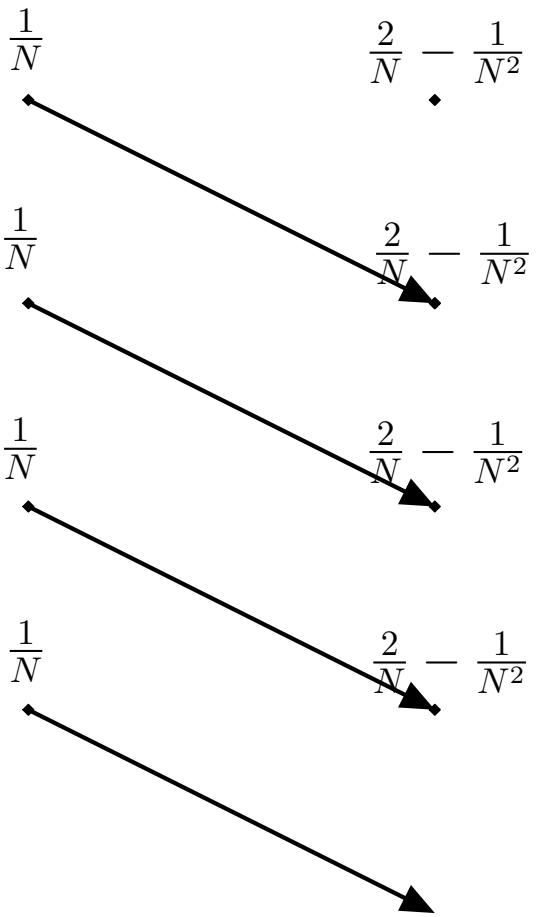
$\rightarrow$ accumulate packets during this phase

Counter Example (second phase)



Second phase of length $(1-\alpha)T$
No collision

Counter Example (second phase)



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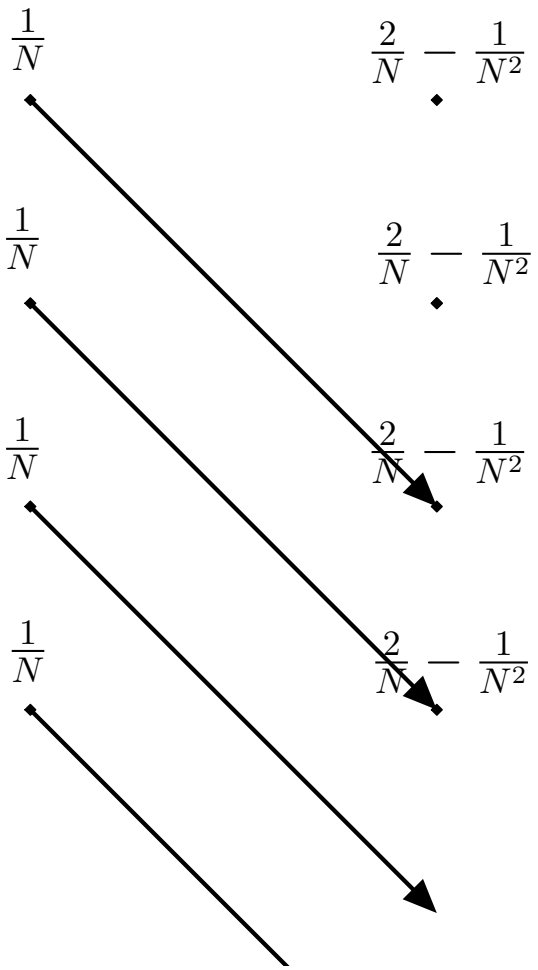
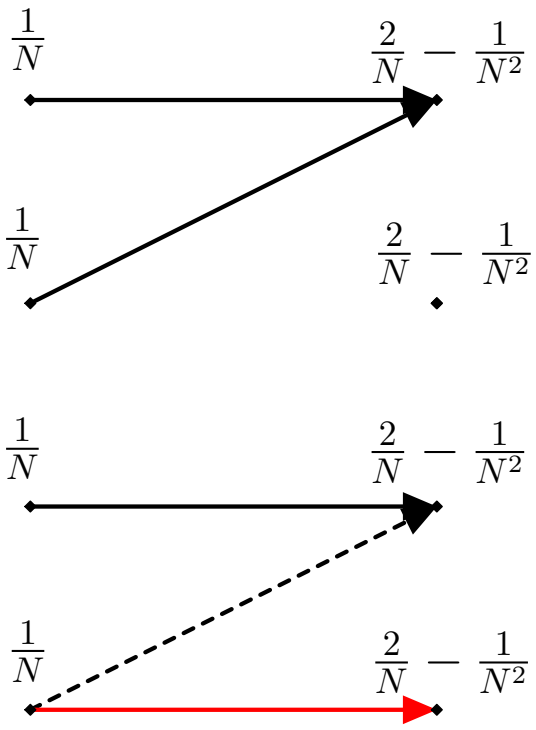


Second phase of length $(1-\alpha)T$
No collision

$\rightarrow$ clear packets during this phase
if α large enough, still accumulate overall $\Omega(T)$ packets
$\rightarrow$ unstable

Counter example (no policy regret)

What if queue i deviates and plays $p \in \mathcal{P}([K])$ at each round?



First phase

$\rightarrow$ clear all packets

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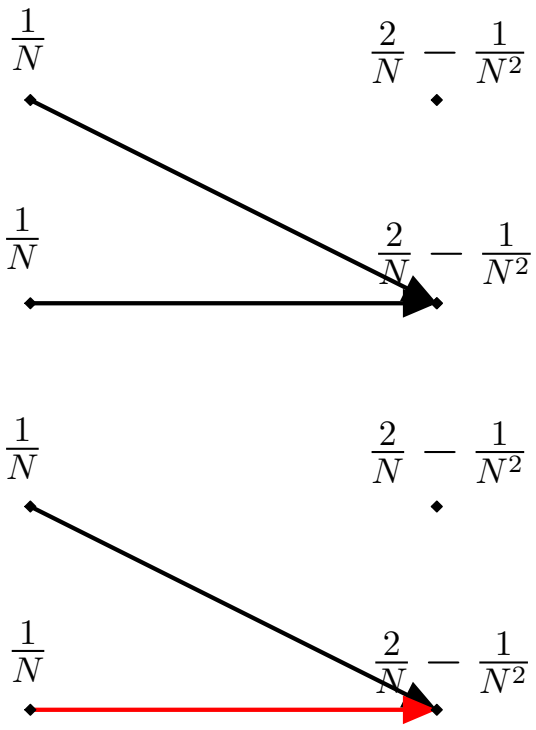

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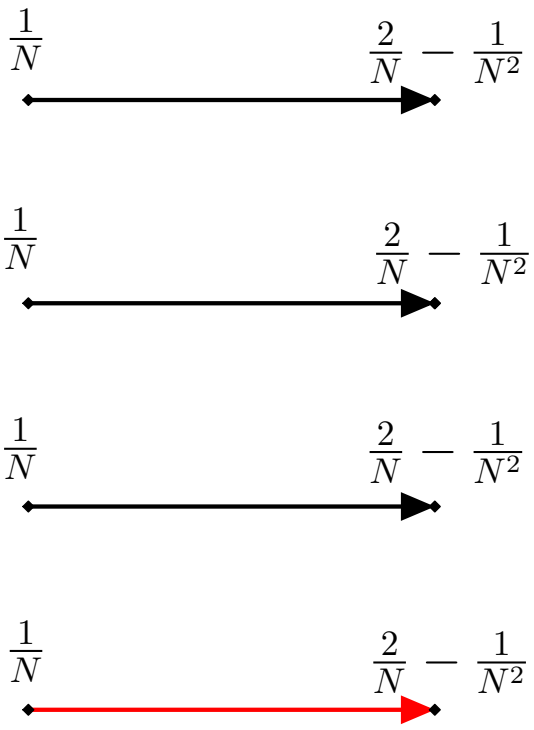

Second phase
many collisions
other queues have priority
accumulate $\Omega(T)$ packets

for α small enough, accumulate more packets when deviating
$\rightarrow$ No policy regret strategies!

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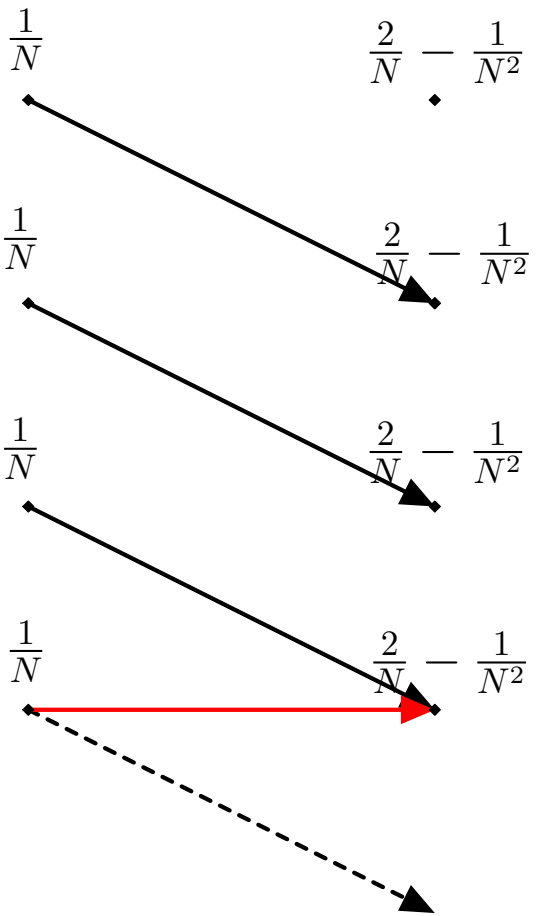

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Case for cooperative algorithm

for some α, strategy profile is both unstable and no policy regret
$\rightarrow$ patience not enough for learning agents with $\eta < 2$
Counter example shows cooperation/coordination is also required

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Centralized strategy when $\eta > 1$: play same joint distribution at each round
$\rightarrow$ correlated equilibria of $\mathcal{G}$
$\rightarrow$ *"correlated price of stability"* $= 1$

Design cooperative learning strategies!

Decentralized learning strategy

A stable learning strategy

Assumptions:

- all queues know N and have attributed ranks $i \in [N]$
- shared randomness between queues

Theorem

If $\eta > 1$ and all queues follow ADeQuA, *then the system is stable.*

Inspired by ε-greedy policy

```
for t = 1, ..., ∞ do
    Draw ω_1 ~ Bernoulli(ε_t)          // shared randomness
    if ω_1 = 1 then Explore
    else Exploit
end
```

ADeQuA: A DEcentralized QUeuing Algorithm

Exploration

All queues explore simultaneously and explore either $\boldsymbol{\mu}$ or $\boldsymbol{\lambda}$

Explore $\boldsymbol{\mu}$: queues choose servers without colliding
$\rightarrow$ accurate estimations of all μ_k

Assumption: servers break ties in packets' age uniformly at random

Explore $\boldsymbol{\lambda}$: when queue i explores queue j, both choose same server k **with packet generated at t (if it exists)**
i clears with probability $(1 - \frac{\lambda_j}{2})\mu_k \rightarrow$ estimate λ_j

Exploitation: notations

$\mathfrak{B}_K =$ set of bistochastic matrices
$\mathfrak{S}_K =$ set of permutations (or permutation matrices) on $[K]$

Dominant mapping $\phi : \mathbb{R}^N \times \mathbb{R}^K \to \mathfrak{B}_K$ such that for any (λ, μ):

$$\lambda < P\mu \qquad \text{if possible}$$

! **Birkhoff von Neumann decomposition** $\psi : \mathfrak{B}_K \to \mathcal{P}(\mathfrak{S}_K)$ such that for any P, $\mathbb{E}[\psi(P)] = P$

For convenience, $\psi(P) : [0, 1] \to \mathfrak{S}_K$

Exploitation: centralized

When centralized:

- $\phi : (\hat{\lambda}, \hat{\mu}) \mapsto$ marginals ensuring stability
- $\psi : P \mapsto$ coupling without collision

Draw $\omega_2 \sim \mathcal{U}(0,1)$	`// shared randomness`
Play $\psi(\phi(\hat{\lambda}, \hat{\mu}))(\omega_2)$	

Centralized exploitation

When decentralized:

- compute mapping $\hat{A}^i = \psi(\phi(\hat{\lambda}^i, \hat{\mu}^i))$
- play $\hat{A}^i(\omega_2)(i)$

Exploitation: need for regularity

$\hat{A}^i$ computed from own estimates $(\hat{\lambda}^i, \hat{\mu}^i)$, which differ (but are close)
General dominant mappings and BvN decompositions are non-continuous

$\|\hat{A}^i - \hat{A}^j\|$ arbitrarily large $\implies$ too many collisions

If ϕ and ψ **regular** $\rightarrow \|\hat{A}^i - \hat{A}^j\|$ small
$\implies$ small amount of collisions

Goal: Design **regular** dominant mapping and BvN decomposition

Dominant mapping

Goal $\phi : \mathbb{R}^N \times \mathbb{R}^K \to \mathfrak{B}_K$ such that for any (λ, μ):

$$\lambda < P\mu \qquad \text{if possible}$$

Usual dominant mappings sort λ and μ $\to$ discontinuity

$$\phi(\lambda, \mu) = \arg\min_{P \in \mathfrak{B}_K} \max_{i \in [N]} - \ln \left(\sum_{j=1}^{K} P_{i,j}\mu_j - \lambda_i \right) + \frac{1}{2K} \|P\|_2^2.$$

- locally Lipschitz objective
- strong convexity $\implies$ regularity of arg min
- optimization methods to approximate ϕ

Dominant mapping: regularity

Define margin $\Delta := \min_{k \in [N]} \frac{1}{k} \sum_{i=1}^{k} (\mu_{(i)} - \lambda_{(i)})$

$$\Delta > 0 \iff \eta > 1$$

Regularity of ϕ

If $\|(\hat{\lambda} - \lambda, \hat{\mu} - \mu)\|_\infty \lesssim \Delta$:

$$\|\phi(\hat{\lambda}, \hat{\mu}) - \phi(\lambda, \mu)\|_2 \lesssim \frac{K}{\Delta} \|(\hat{\lambda} - \lambda, \hat{\mu} - \mu)\|_\infty,$$

BvN decomposition

input : $\hat{P} \in \mathfrak{B}_K$
$j \leftarrow 1$
$G \leftarrow$ complete $K \times K$ bipartite graph
while $\hat{P} \neq \mathbf{0}$ **do**
  For all $(i,k) \notin \mathrm{supp}(\hat{P})$, remove edge (i,k) in G
  $P[j] \leftarrow$ perfect matching in G `// Hopcroft-Karp algorithm`
  $z[j] \leftarrow \min_{(i,k)\in \mathrm{supp}(P[j])} \hat{P}_{i,k}$ `// maximal weight we can remove`
  $\hat{P} \leftarrow \hat{P} - z[j]P[j]$ and $j \leftarrow j+1$
end
return $(z[j], P[j])_j$

Birkhoff algorithm

BvN decomposition of $\hat{P}$ (non-continuous)
computationally efficient: sequence of length $\leq K^2$

$$\psi(P)(\omega) = P[j_\omega]$$
where $\sum_{j \leq j_\omega} z[j] \leq \omega < \sum_{j \leq j_\omega + 1} z[j]$.

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BvN decomposition

input : $\hat{P} \in \mathfrak{B}_K$, $C \in \mathbb{R}^{K \times K}$ (cost matrix)
$j \leftarrow 1$
while $\hat{P} \neq \mathbf{0}$ **do**
  For all $(i,k) \notin \mathrm{supp}(\hat{P})$, $C_{i,k} \leftarrow +\infty$ `// remove edge` (i,k)
  $P[j] \leftarrow \mathrm{Hungarian}(C)$ `// matching with minimal cost w.r.t.` C
  $z[j] \leftarrow \min_{(i,k) \in \mathrm{supp}(P[j])} \hat{P}_{i,k}$
  $\hat{P} \leftarrow \hat{P} - z[j]P[j]$ and $j \leftarrow j+1$
end
return $(z[j], P[j])_j$

Ordered Birkhoff algorithm

return $P[j]$ in increasing order

$$P[j] \prec_C P[j+1] \iff \sum_{i,k} C_{i,k} P[j]_{i,k} < \sum_{i,k} C_{i,k} P[j+1]_{i,k}$$

$$\psi(P)(\omega) = P[j_\omega]$$

where $\sum_{j \leq j_\omega} z[j] \leq \omega < \sum_{j \leq j_\omega + 1} z[j]$.

BvN decomposition: regularity

Proposition:
$z[j]$ = the j-th non-zero of sequence $(z_l(P))_{1 \leq l \leq K!}$ with

$$z_j(P) = \min_{(i,k) \in \mathrm{supp}(P_j)} \left(P - \sum_{l=1}^{j-1} z_l(P) P_l \right)_{i,k} \tag{1}$$

where (P_j) increasing sequence of permutation matrices

$\rightarrow$ continuity by induction

Regularity of ψ

$$\underbrace{\mathbb{P}_{\mathcal{U}(0,1)}(\psi(\hat{P}^i) \neq \psi(\hat{P}^j))}_{\geq \text{ probability of collision}} \leq 2^{2K^2} \|\hat{P}^i - \hat{P}^j\|_\infty.$$

Stability result

Take $\varepsilon_t = (N+K)t^{-\frac{1}{5}}$

Theorem (Stability)

If $\eta > 1$, for any queue i and any $r \in \mathbb{N}$, there exists a constant C_r depending only on r such that

$$\mathbb{E}[(Q_t^i)^r] \leq C_r K N \left(\frac{N^{\frac{5}{2}} K^{\frac{5}{2}} 2^{5K^2}}{(\min(1, K\bar{\mu}) \min_i \lambda_i)^{\frac{5}{4}} \Delta^5} \right)^r, \quad \textit{for all } t \in \mathbb{N}.$$

- stability with $\eta > 1$
- super exponential dependency due to BvN decomposition
- asymptotic number of packets similar to centralized case

Sketch of Proof

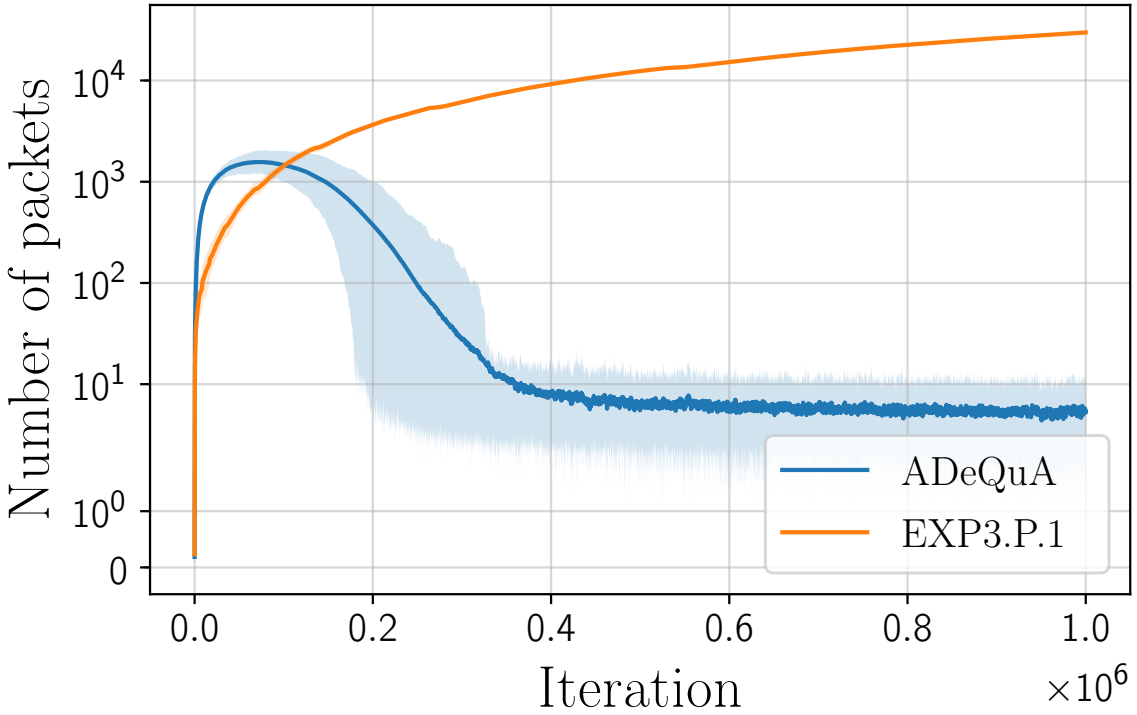
For $t \geq \overbrace{\tau}^{\text{learning time}}$, λ and μ are estimated tightly enough (concentration inequalities)

Afterwards, regularity of ϕ and ψ lead to

- small number of collisions
- queues clear with proba $\lambda_i + \mathcal{O}(\Delta)$

$\implies$ number of packets decreases after τ

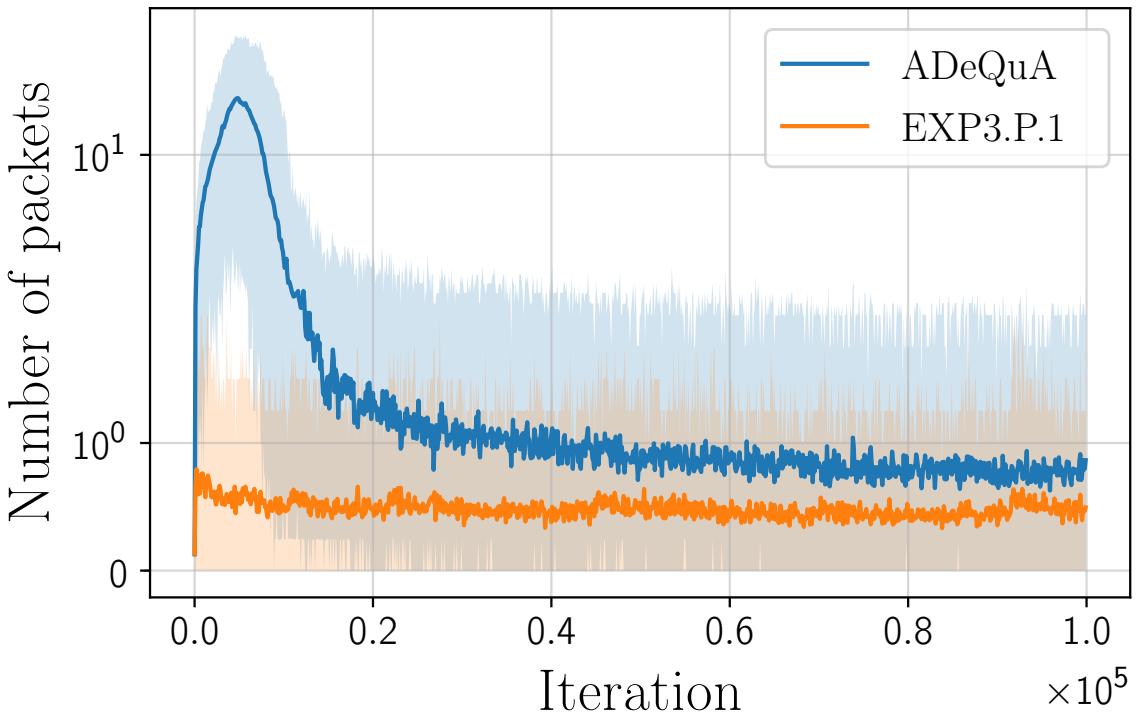
Simulations: hard instance



Hard instance, $\eta < 2$.

- No regret strategies: unstable
- ADeQuA: stable
  + number of packets decreases after learning

Simulations: easy instance



Easy instance, $\eta > 2$.

- both strategies stable
- No regret better suited to easy instances?

Recap

- carryover effect incites cooperation in queuing systems
- patience not enough for learning
- first decentralized stable learning strategy
- confirmed empirically

Open questions

- design of better dominant mapping and BvN decomposition
- dynamic case: queues don't have a common time clock
- enhancement with additional observation (eg collision information)

Thanks!